

Statistics – The Backbone of AI/ML

Linear Algebra – The Backbone

TOPIC :

Introduction to statistics

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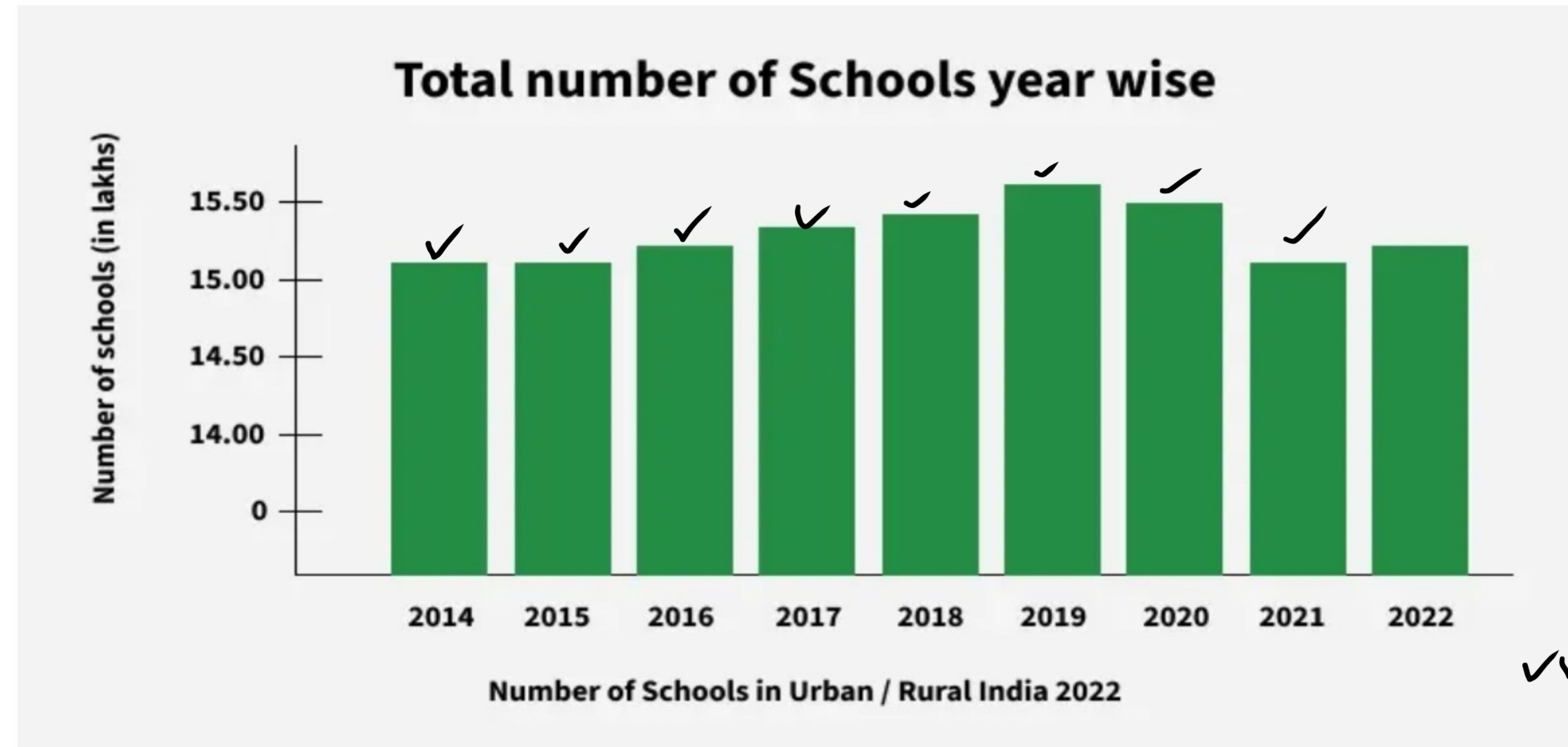


CONTENT

- 1) Statistics
 - ↓
- 2) Types of Statistics
 - Descriptive ——— Mean, Median, Mode, Variance, SD
 - Inference
 - Probability Distribution
 - Hypothesis Testing (p-value, etc)
t-value
 - etc (Histogram, Bar chart, Pie chart)
- 3) Mean
 - A.M
 - H.M
 - G.M



Example.



Statistics

Statistics is a Predicting and Analysing data!

Data — collecting

↓
Manipulate

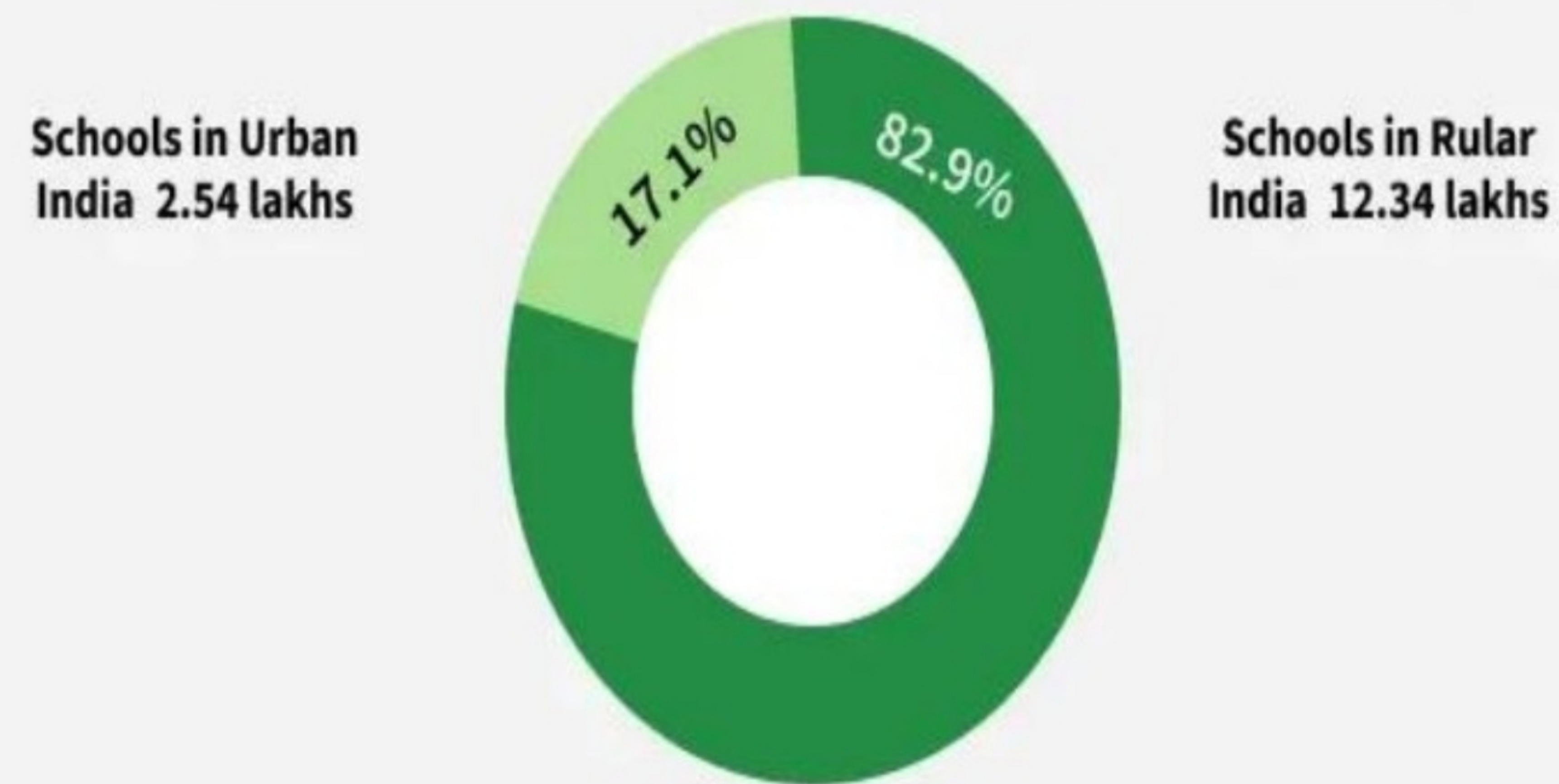
↓
Analyse

Data :- Raw fact or collection of information -



Ex-2

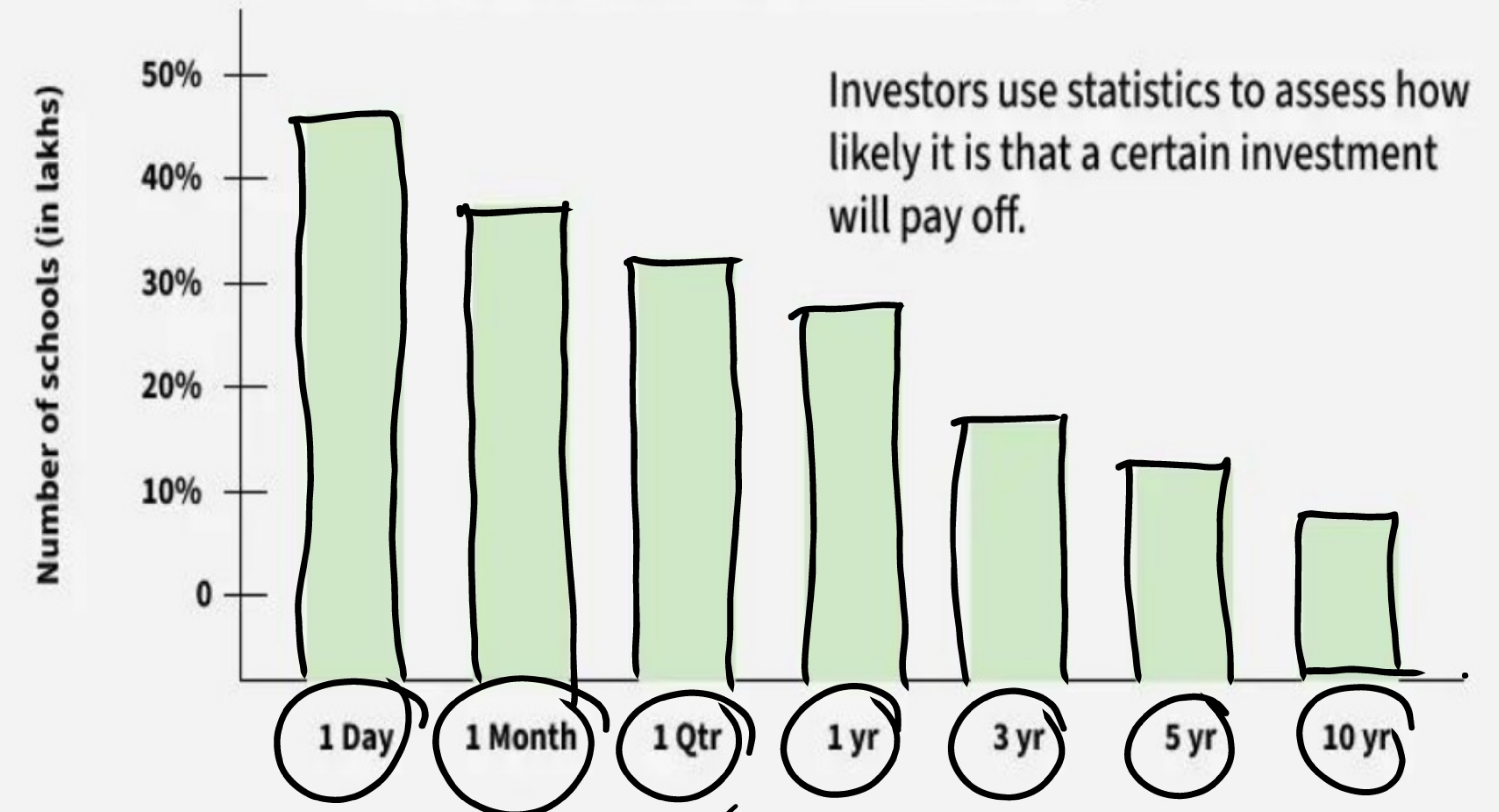
Number of Schools in Urban/ Rural India



the increase in the number of schools in India in the past few years.

Ex-3

Probabilities of Stock Investing



(Bar-diagram)

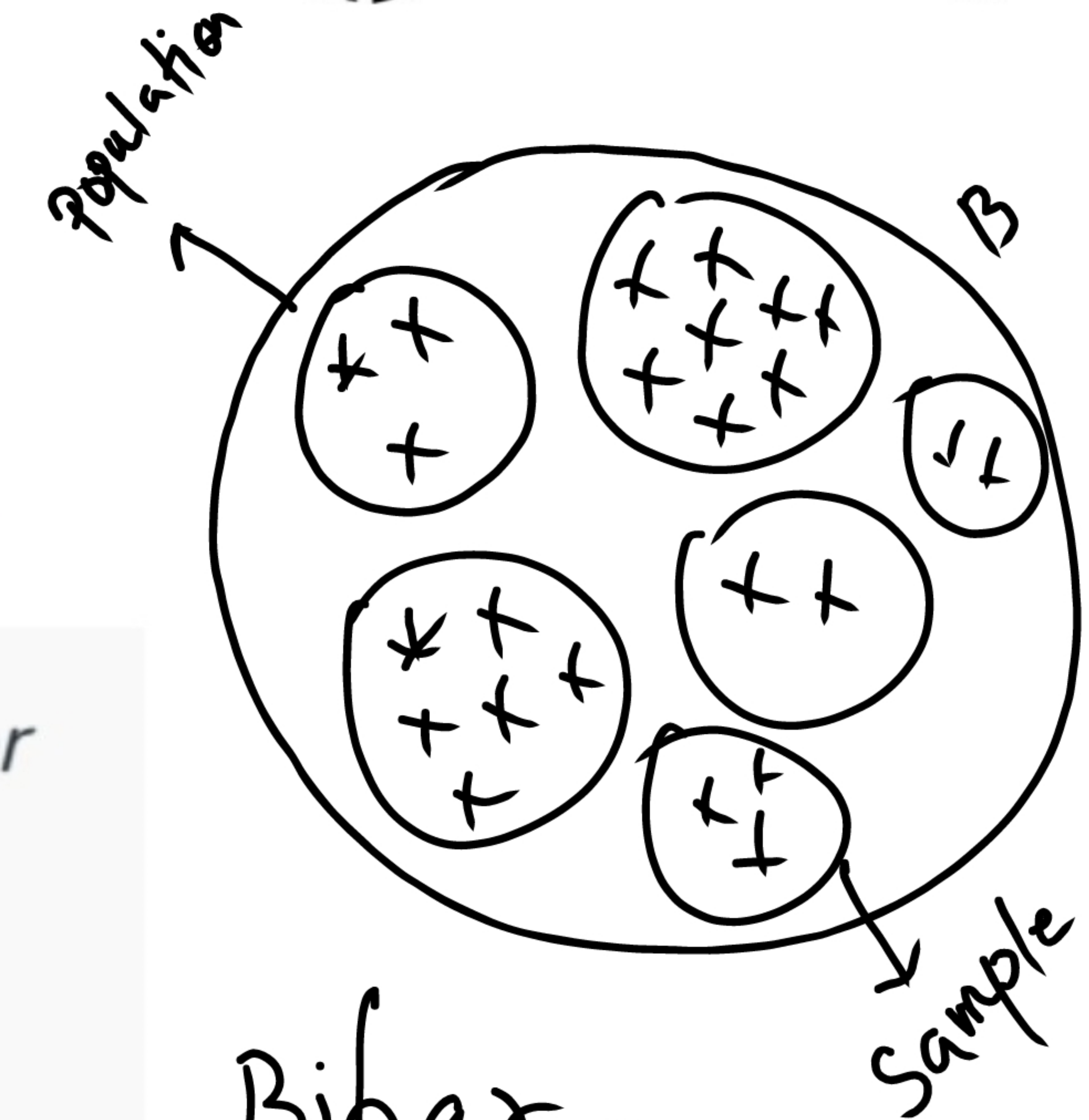
Data - Analyse → Report → Statistics ✓



Statistics Terminologies

Some of the most common terms you might come across in statistics are:

- **Population:** It is actually a collection of a set of individual objects or events whose properties are to be analyzed.
- **Sample:** It is the subset of a population.
- **Variable:** It is a characteristic that can have different values.
- **Parameter:** It is numerical characteristic of population.



Bihar -
Election poll
area wise
BJP / NDA / RJD etc



Types of Statistics

Statistics is the study of data; when we talk about properties of data, it comes under descriptive statistics. Meanwhile, the study of concluding data comes under inferential statistics.

$$\text{Marks} = \left\{ \begin{array}{c} \text{20bacca} \\ 70\%, 95\%, 80\%, 75\% \dots \end{array} \right\}$$

$$\text{Average} = \left(\frac{\sum x}{N} \right)$$

Statistics is the foundation of data science. Descriptive statistics are simple tools that help us understand and summarize data. They show the basic features of a dataset, like the average, highest and lowest values and how spread out the numbers are. It's the first step in making sense of information.

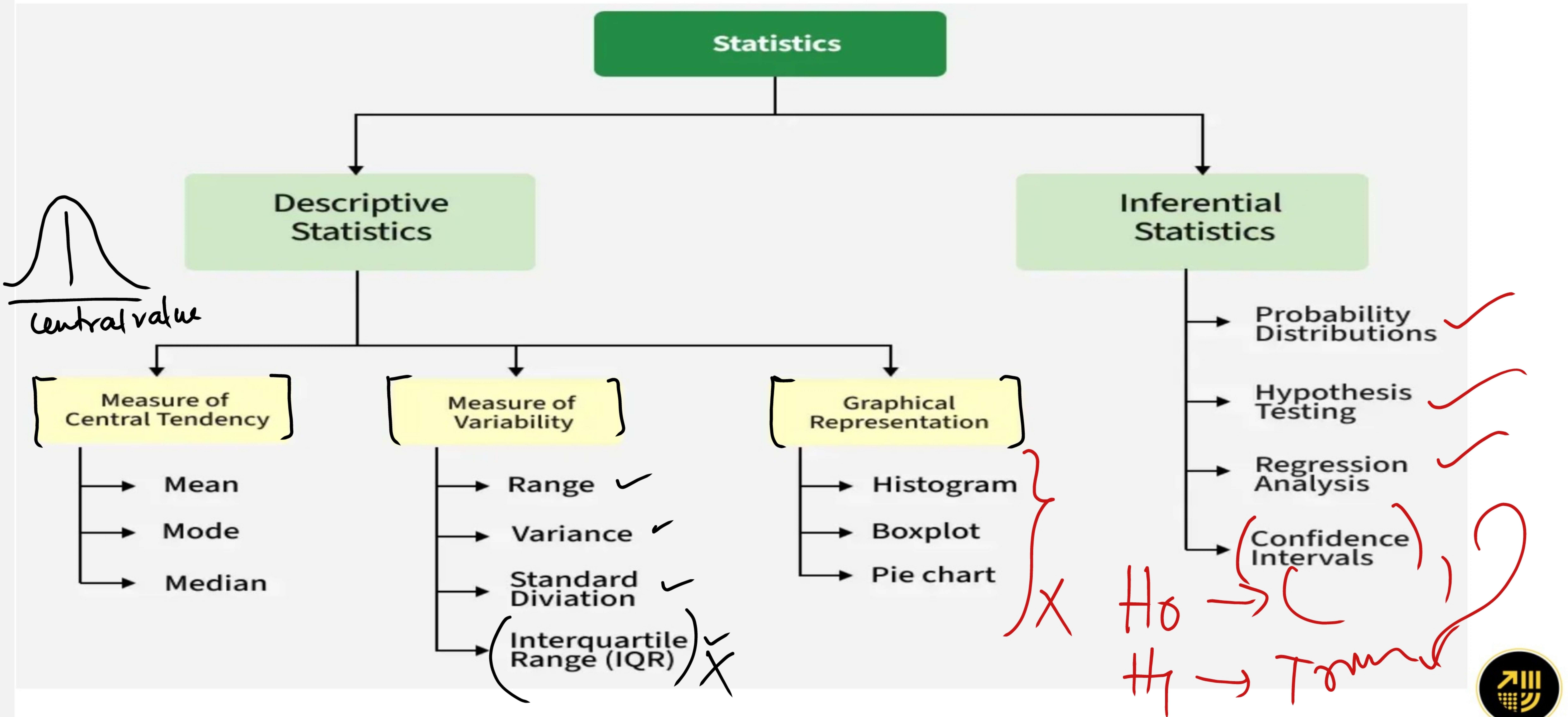
Descriptive

Mean, Media, Mode, Variance, SD

$\int f(x)dx = \textcircled{1}$ Inference

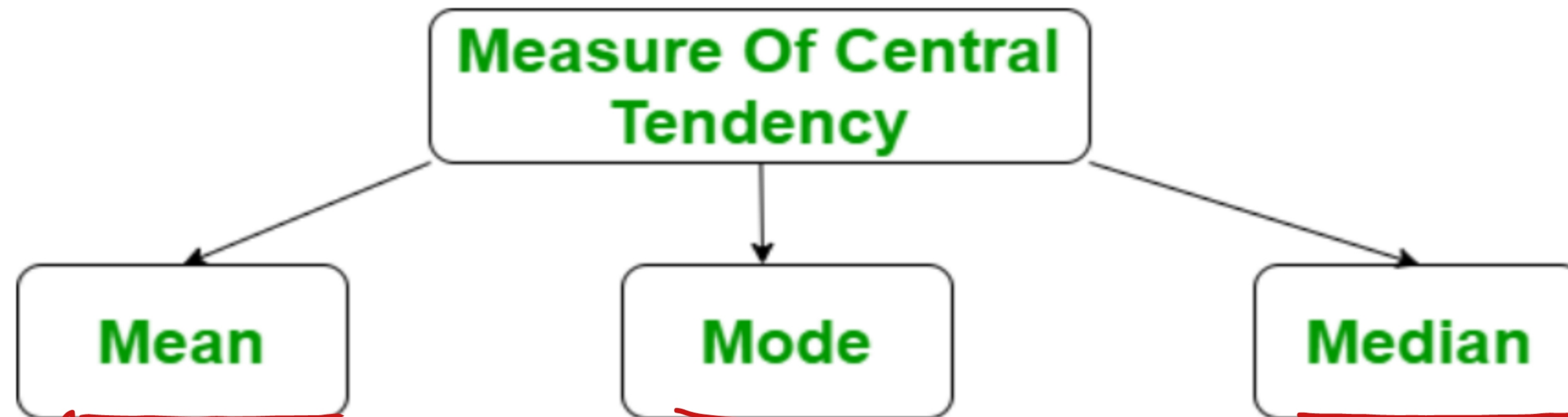
Probability distributio, Hypothesis Testing.





The representative value of a data set, generally the central value or the most occurring value that gives a general idea of the whole data set is called Measure of Central Tendency.

Statistical values that describe the central position within a dataset. There are three main measures of central tendency:



Measures of Central Tendency

$$\sum \frac{x}{N}$$

Frequency

$$S = \{1, 2, 2, \underline{3, 3, 3}, 4, 5\} = 3$$

Midvalue
 ()

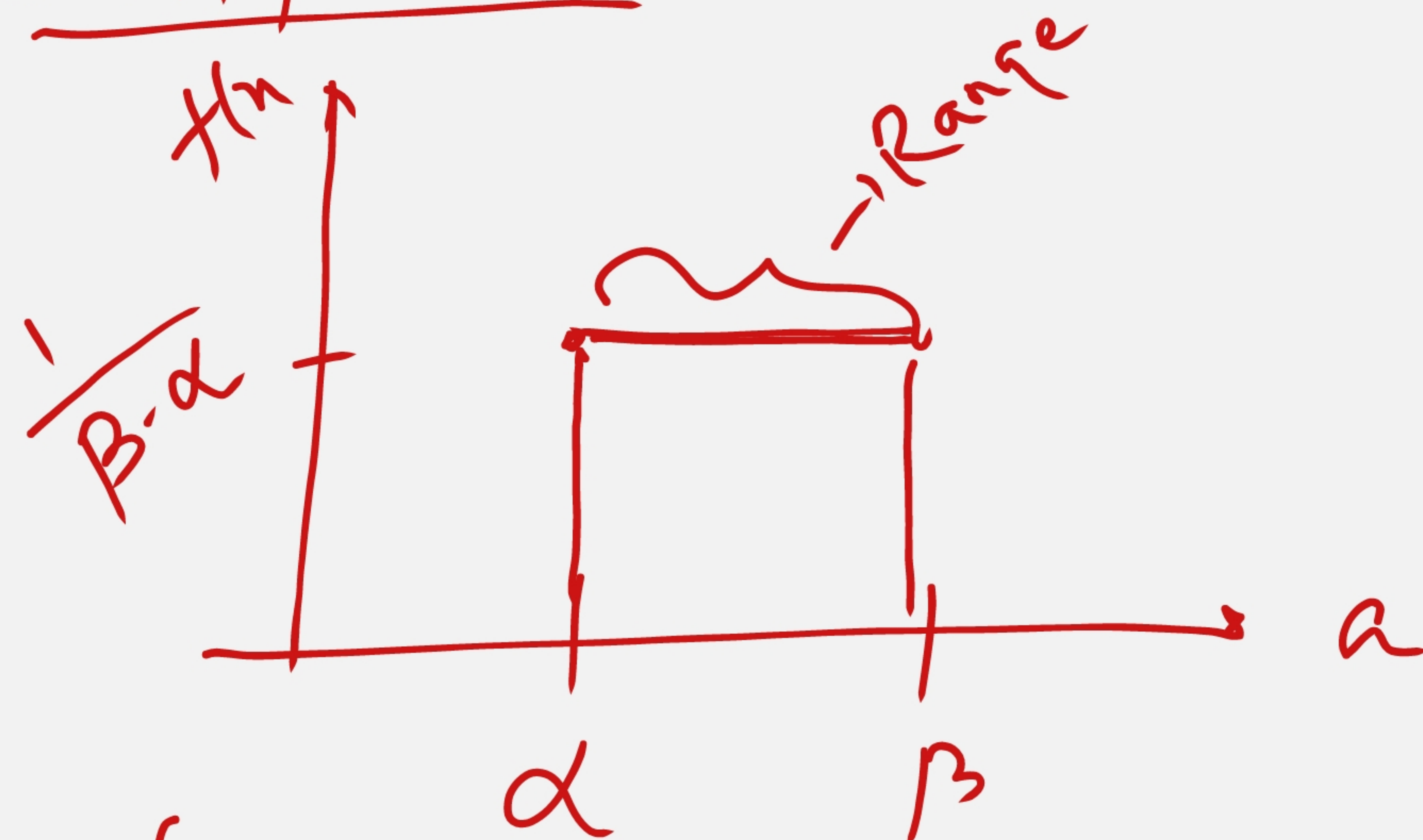


Uniform distribution

In a general we say that X is a uniform random variable on the interval (a, b) if its probability density func is given by -

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha < x < \beta \\ 0 & , \text{ otherwise} \end{cases}$$

Graphical Representation



For uniform distribution -

$$\text{Mean} (E(x)) = \frac{\beta + \alpha}{2}$$

$$\text{Variance} = V(x) = \frac{(\beta - \alpha)^2}{12}$$

Proof
for both

5 - 50

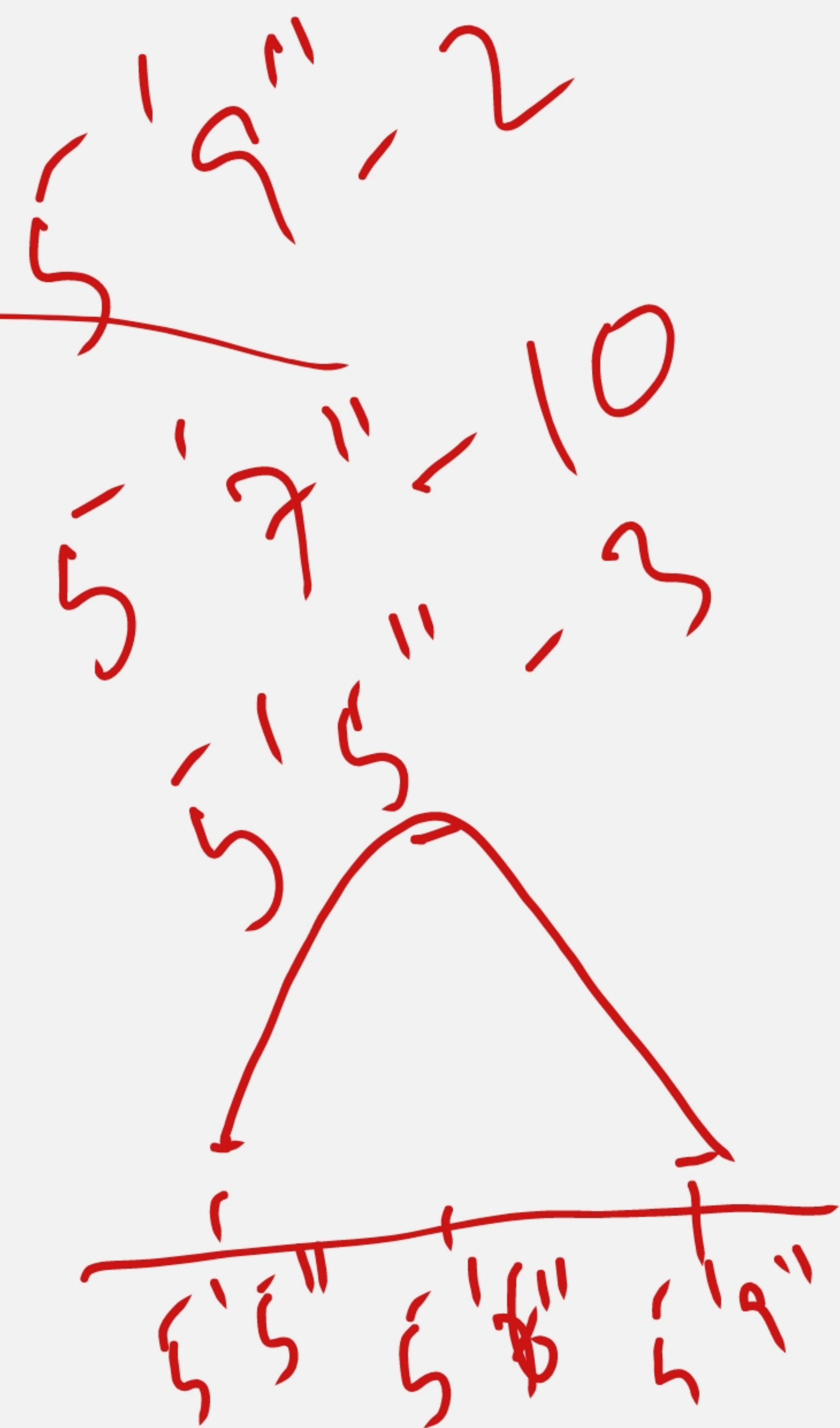
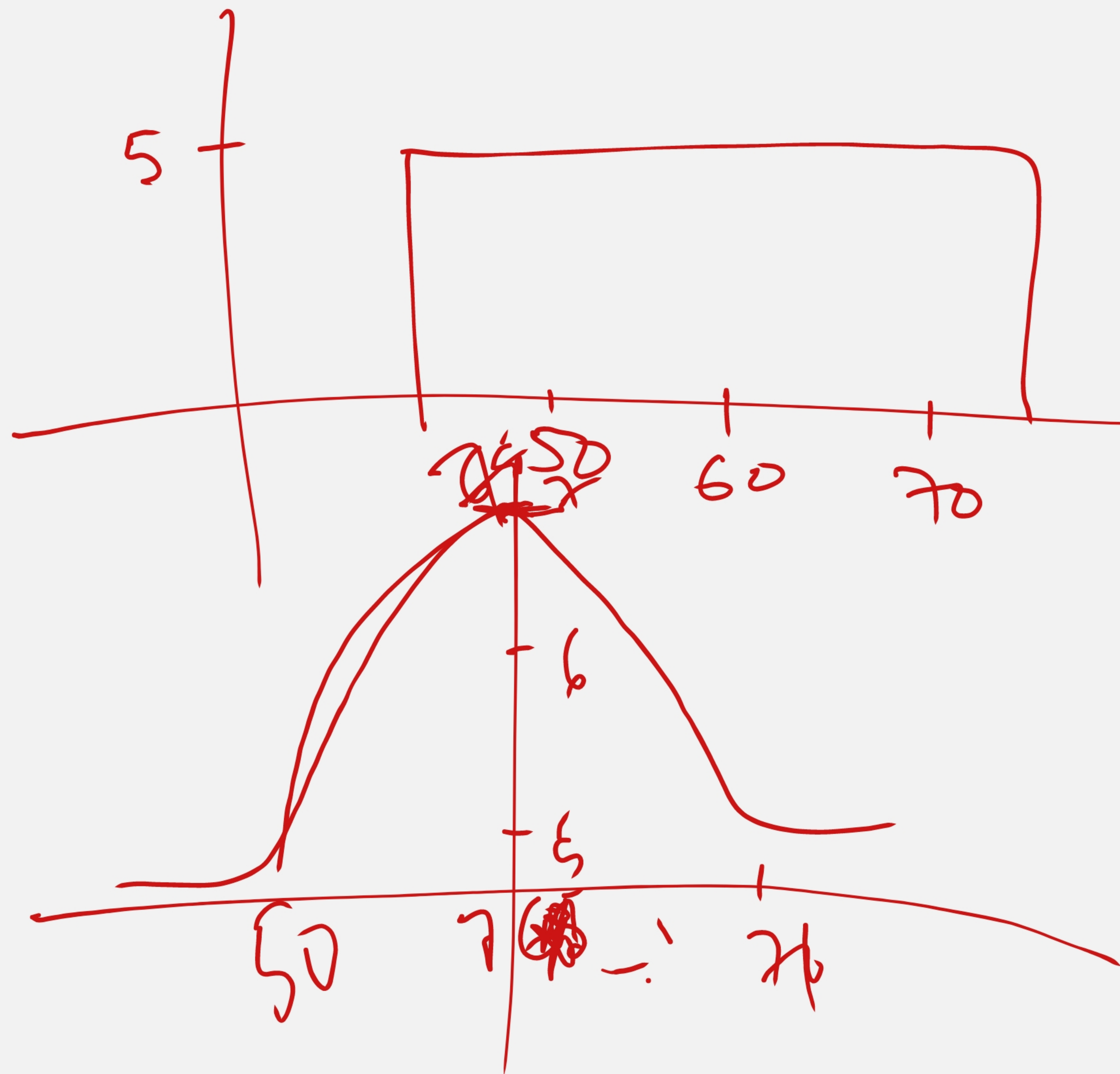
5 - 60

5 - 70

5 - 50

7 - 65

5 - 71



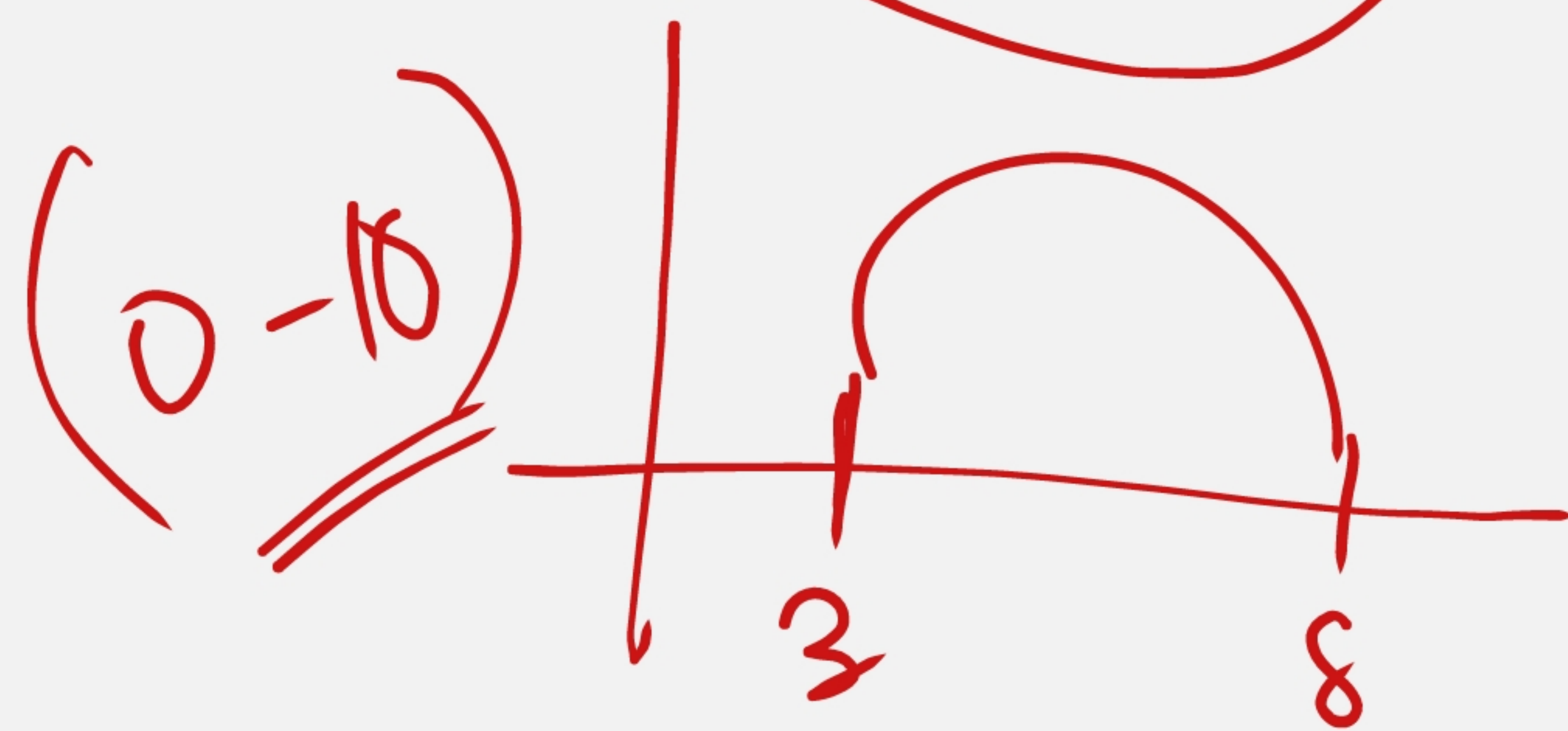
Problem

If X is Uniformly distributed over $(0, 10)$
Calculate the probability that -

(a) $X < 3$ (0 to 3)

(b) $X > 6$ (6 to 10)

(c) $3 < X < 8$

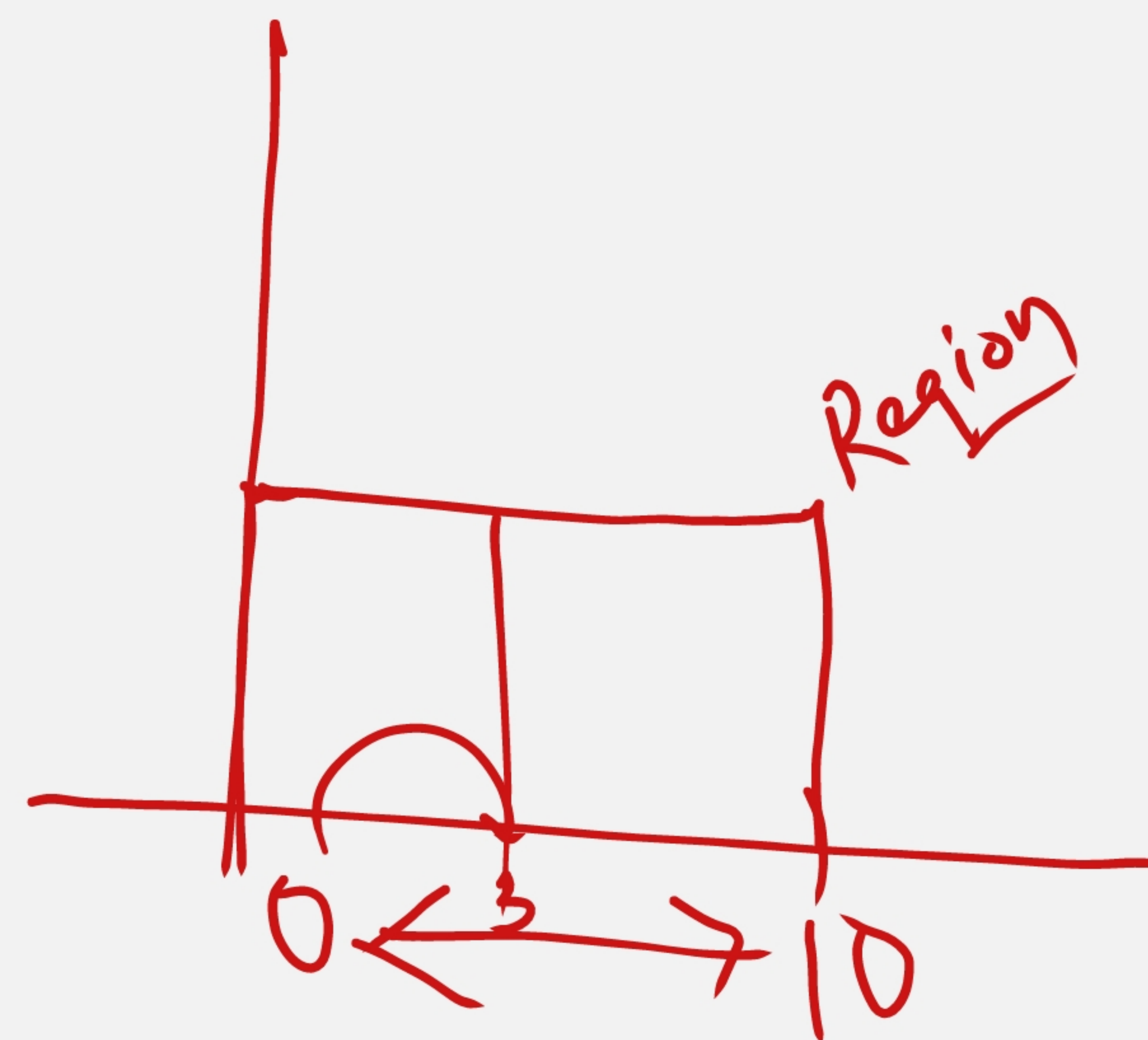


$$f(x) = \frac{1}{10 - 0} = \frac{1}{10}$$

(a) $P(X < 3) = \int_0^3 \frac{1}{10} dx = \frac{3}{10}$ a

(b) $P(X > 6) = \int_6^{10} \frac{1}{10} dx = \frac{4}{10} = \frac{2}{5}$

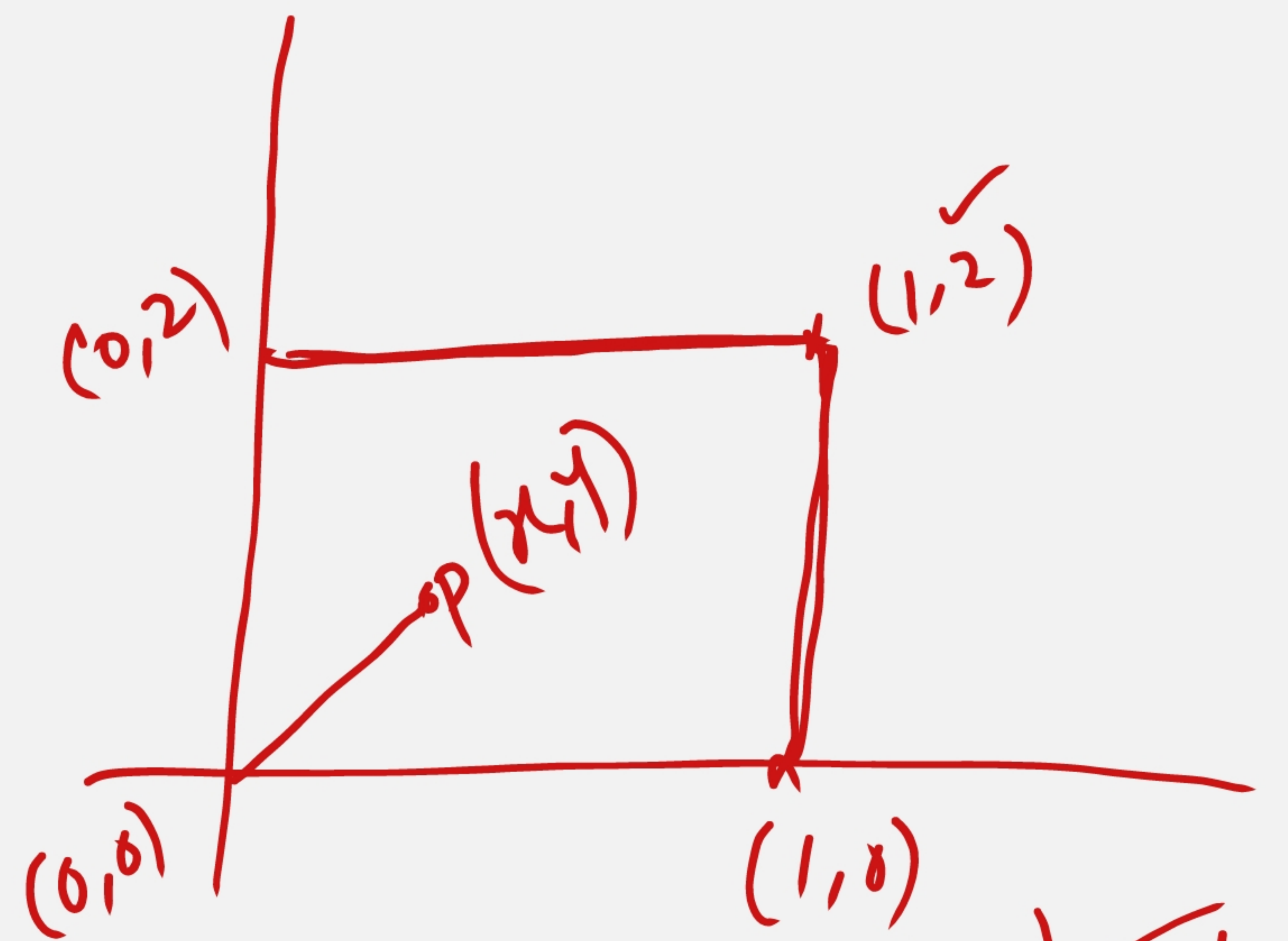
(c) $P(3 < X < 8) = \int_3^8 \frac{1}{10} dx = \frac{1}{2}$



$$\frac{1}{b - a}$$

problem

A point is randomly selected with uniform probability in the x - y plane within the rectangle with corners at $(0,0)$, $(1,0)$, $(1,2)$ and $(0,2)$. If p is the length of the position vector of the point, the expected value of $E(p^2) = ?$



p^2 is -

- (a) $2/3$ (b) 1 (c) $4/3$ (d) $5/3$

$$\begin{aligned} \checkmark \text{pdf} &= \frac{1}{1-0} = 1 \\ \checkmark \text{pdf} &= \frac{1}{2-0} = \frac{1}{2} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \frac{1}{b-a}$$

Speciality { length of position vector of point $p = \sqrt{(x-0)^2 + (y-0)^2} = \sqrt{x^2 + y^2}$

$$\begin{aligned} \checkmark p^2 &= x^2 + y^2 \\ \checkmark E(p^2) &= E(x^2) + E(y^2) \end{aligned}$$

$$E(X^2) = \int_0^1 x^2 p(x) dx = \int_0^1 x^2 - 1 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}.$$

$$E(Y^2) = \int_0^2 y^2 p(x) dx = \int_0^2 y^2 - \frac{1}{2} dx = \left[\frac{y^3}{6} \right]_0^2 = \frac{8}{6} = \frac{4}{3}.$$

$$\begin{aligned} \therefore E(P^2) &= E(X^2) + E(Y^2) \\ &= \frac{1}{3} + \frac{4}{3} = \frac{5}{3} \quad \checkmark \end{aligned}$$